JAMA | Original Investigation

Early Intervention in Patients With Asymptomatic Severe Aortic Stenosis and Myocardial Fibrosis

The EVOLVED Randomized Clinical Trial

Krithika Loganath, MD; Neil J. Craig, MD; Russell J. Everett, PhD; Rong Bing, PhD; Vasiliki Tsampasian, MD; Patrycja Molek, MD; Simona Botezatu, MD; Saadia Aslam, MD; Steff Lewis, PhD; Catriona Graham, MSc; Audrey C. White; Tom MacGillivray; Christopher E. Tuck; Phillip Rayson, (BA)Hons; Denise Cranley; Sian Irvine, PhD; Ruth Armstrong; Lynsey Milne; Calvin W. L. Chin, PhD; Graham S. Hillis, PhD; Timothy Fairbairn, PhD; John P. Greenwood, PhD; Richard Steeds, PhD; Stephen J. Leslie, PhD; Chim C. Lang, PhD; Chiara Bucciarelli-Ducci, PhD; Nikhil V. Joshi, PhD; Vijay Kunadian, PhD; Vassilios S. Vassiliou, PhD; Jason N. Dungu, PhD; Sandeep S. Hothi, PhD; Nicholas Boon, PhD; Sanjay K. Prasad, PhD; Niall G. Keenan, MD; Dana Dawson, PhD; Thomas A. Treibel, PhD; Mani Motwani, PhD; Christopher A. Miller, PhD; Nicholas L. Mills, PhD; Ronak Rajani, PhD; David P. Ripley, PhD; Gerry P. McCann, MD; Bernard Prendergast, MD; Anvesha Singh, PhD; David E. Newby, MD; Marc R. Dweck, PhD; for the EVOLVED investigators

IMPORTANCE Development of myocardial fibrosis in patients with aortic stenosis precedes left ventricular decompensation and is associated with an adverse long-term prognosis.

OBJECTIVE To investigate whether early valve intervention reduced the incidence of all-cause death or unplanned aortic stenosis–related hospitalization in asymptomatic patients with severe aortic stenosis and myocardial fibrosis.

DESIGN, SETTING, AND PARTICIPANTS This prospective, randomized, open-label, masked end point trial was conducted between August 2017 and October 2022 at 24 cardiac centers across the UK and Australia. Asymptomatic patients with severe aortic stenosis and myocardial fibrosis were included. The final date of follow-up was July 26, 2024

INTERVENTION Early valve intervention with transcatheter or surgical aortic valve replacement or guideline-directed conservative management.

MAIN OUTCOMES AND MEASURES The primary outcome was a composite of all-cause death or unplanned aortic stenosis–related hospitalization in a time-to-first-event intention-to-treat analysis. There were 9 secondary outcomes, including the components of the primary outcome and symptom status at 12 months.

RESULTS The trial enrolled 224 eligible patients (mean [SD] age, 73 [9] years; 63 women [28%]; mean [SD] aortic valve peak velocity of 4.3 [0.5] m/s) of the originally planned sample size of 356 patients. The primary end point occurred in 20 of 113 patients (18%) in the early intervention group and 25 of 111 patients (23%) in the guideline-directed conservative management group (hazard ratio, 0.79 [95% CI, 0.44-1.43]; $P = .44$; between-group difference, −4.82% [95% CI, −15.31% to 5.66%]). Of 9 prespecified secondary end points, 7 showed no significant difference. All-cause death occurred in 16 of 113 patients (14%) in the early intervention group and 14 of 111 (13%) in the guideline-directed group (hazard ratio, 1.22 [95% CI, 0.59-2.51]) and unplanned aortic stenosis hospitalization occurred in 7 of 113 patients (6%) and 19 of 111 patients (17%), respectively (hazard ratio, 0.37 [95% CI, 0.16-0.88]). Early intervention was associated with a lower 12-month rate of New York Heart Association class II-IV symptoms than guideline-directed conservative management (21 [19.7%] vs 39 [37.9%]; odds ratio, 0.37 [95% CI, 0.20-0.70]).

CONCLUSIONS AND RELEVANCE In asymptomatic patients with severe aortic stenosis and myocardial fibrosis, early aortic valve intervention had no demonstrable effect on all-cause death or unplanned aortic stenosis–related hospitalization. The trial had a wide 95% CI around the primary end point, with further research needed to confirm these findings.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: NCT03094143

Visual Abstract
Editorial
Supplemental content

JAMA. doi:10.1001/jama.2024.22730
Published online October 28, 2024.

Author Affiliations: Author affiliations are listed at the end of this article.

Corresponding Author: Marc R. Dweck, PhD, University of Edinburgh, 49 Little France Crescent, Edinburgh EH16 4SB, United Kingdom (marc.dweck@ed.ac.uk).

© 2024 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Michael Bowdish on 10/30/2024

Aortic stenosis is the most common heart valve disease in high-resource countries, with an increasing prevalence in the aging population.[1] Aortic valve replacement via surgical or transcatheter approaches remains the cornerstone of treatment, but is reserved for patients with severe aortic stenosis who are symptomatic or those with a left ventricular ejection fraction below 50%.[2,3] Based on expert opinion and nonrandomized data, guidelines recommend that asymptomatic patients are observed and that aortic valve intervention is deferred until the onset of symptoms. In clinical practice, assessment of symptoms in patients with severe aortic stenosis is challenging due to limited mobility or multiple comorbidities.[4,5] Two small randomized clinical trials have suggested that early surgical aortic valve replacement may improve clinical outcomes in select younger patients with asymptomatic severe aortic stenosis and normal left ventricular ejection fraction.[6,7]

The potential benefits of early aortic valve intervention are most likely to be apparent in patients who are at the highest risk of aortic stenosis–related clinical events. In patients with aortic stenosis, plasma high-sensitivity cardiac troponin I concentration and left ventricular hypertrophy on electrocardiography are sensitive markers of myocardial health and adverse left ventricular remodeling that are associated with worse outcomes.[8-11] Midwall late gadolinium enhancement on cardiac magnetic resonance imaging provides a more definitive specific measure of cardiac damage through the identification of myocardial fibrosis, the key pathological process driving the transition from left ventricular hypertrophy to heart failure in aortic stenosis.[12-19] Multiple observational studies have demonstrated that myocardial fibrosis progresses rapidly once established and is a strong independent predictor of incident heart failure and all-cause and cardiovascular mortality in patients with aortic stenosis.[20-23] This study therefore developed an enrichment approach using these biomarkers to identify asymptomatic patients with severe aortic stenosis who had evidence of myocardial fibrosis and who would be at heightened risk of cardiac decompensation.

The Early Valve Replacement Guided by Biomarkers of Left Ventricular Decompensation in Asymptomatic Patients with Severe Aortic Stenosis (EVOLVED) trial was designed to investigate whether early aortic valve intervention can improve clinical outcomes in patients with asymptomatic severe aortic stenosis and myocardial fibrosis. It was hypothesized that the incidence of all-cause death or unplanned aortic stenosis–related hospitalization would be reduced in patients who underwent early aortic valve intervention compared with those receiving guideline-directed conservative management.

Key Points

Question Is early aortic valve intervention superior to guideline-directed conservative management in asymptomatic patients with severe aortic stenosis and myocardial fibrosis?

Findings In this multicenter randomized clinical trial of 224 patients with asymptomatic severe aortic stenosis and myocardial fibrosis, there was no significant difference in the primary composite end point of all-cause death or unplanned aortic stenosis–related hospitalization in patients randomized to receive early intervention vs patients randomized to receive guideline-directed conservative management: 18% vs 23%.

Meaning Among patients with asymptomatic severe aortic stenosis and myocardial fibrosis, early aortic valve intervention did not improve clinical outcomes compared with guideline-directed conservative management.

Methods

Trial Design and Oversight

The EVOLVED trial is a parallel-group, multicenter, prospective, randomized, open-label, masked end point trial conducted across 24 cardiac centers in the UK and Australia (Supplement 1).[24] The study was approved by the South-East Scotland Research Ethics Committee. The trial protocol (Supplement 2) was designed by the chief investigator and approved by the principal investigators and institutional review boards at each participating site. All participants provided written informed consent. A trial steering committee oversaw trial conduct and progress, including data and safety monitoring because this was an open-label trial in which the risks and benefits of all trial-related procedures and interventions are well known. This report follows the CONSORT reporting guideline for parallel-group randomized trials.[25]

Participant Selection

Patients aged 18 years or older with severe aortic stenosis and without symptoms attributable to their valve disease were invited to participate. Severe aortic stenosis was defined as aortic valve peak velocity greater than or equal to 4.0 m/s or an aortic valve peak velocity greater than or equal to 3.5 m/s with an indexed aortic valve area less than 0.6 cm^2/m^2.[2,26] The attending physician assessed for the presence of aortic stenosis–related symptoms with the option of exercise stress testing according to their clinical practice. Routine exercise stress testing was not mandated because of the challenges of interpretation and consequent exclusion of many patients who would be unable to perform exercise stress due to poor mobility or comorbidities. Patients were excluded if they had symptoms attributable to aortic stenosis, left ventricular ejection fraction less than 50%, concomitant severe aortic or mitral regurgitation, estimated glomerular filtration rate less than 30 mL/min/1.73 m^2, contraindications to magnetic resonance imaging, or if deemed unfit for surgery or transcatheter aortic valve implantation (eMethods in Supplement 1).

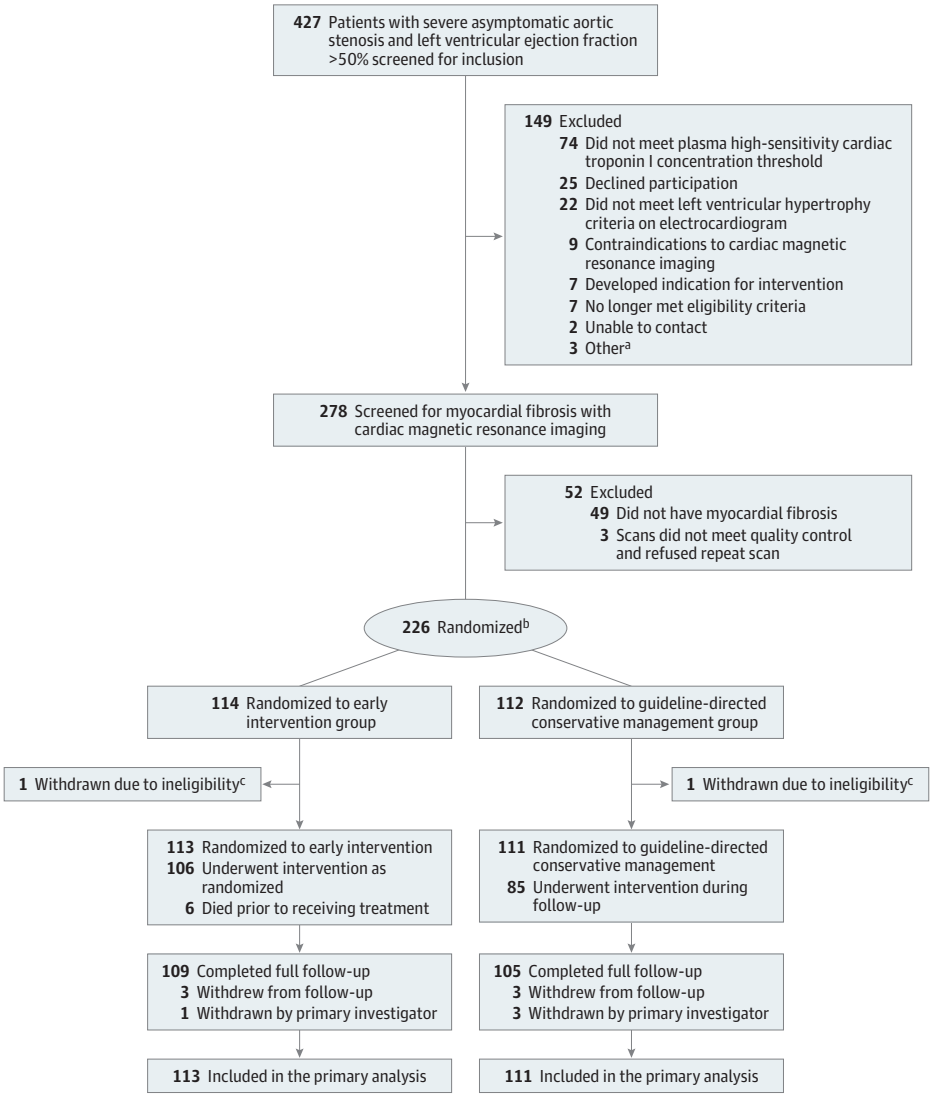
Participant Eligibility

Patients were initially screened for adverse left ventricular remodeling by plasma cardiac troponin I concentration greater than or equal to 6 ng/L measured using a high-sensitivity assay (Abbott Laboratories) or the presence of left ventricular hypertrophy on electrocardiography.[8-11] Potentially eligible participants meeting any of these criteria underwent cardiac magnetic resonance imaging with gadolinium contrast using a standardized protocol (eTable 1 in Supplement 1). The presence or absence of midwall late gadolinium enhancement was assessed by the core

© 2024 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Michael Bowdish on 10/30/2024

Figure 1. Recruitment, Randomization, and Follow-Up in the EVOLVED Trial



[a]Three participants who did not proceed to cardiac magnetic resonance imaging had no clear reason recorded on the trial database.

[b]Randomization used minimization with stratification for age, sex, aortic valve peak velocity, ischemic heart disease, and screening method.

[c]Two participants were excluded because they were randomized after already being referred for surgery (see eMethods in Supplement 1 for details).

laboratory blinded to clinical details (eMethods and eFigure 1 in Supplement 1). Site investigators and attending physicians were blinded to the cardiac magnetic resonance imaging findings other than any unexpected clinically urgent findings.

Randomization

Eligible participants with midwall myocardial fibrosis were randomly assigned in a 1:1 ratio to receive early aortic valve intervention or guideline-directed management using a computer-generated randomization process using minimization incorporating age, sex, aortic valve peak velocity, ischemic heart disease, and screening method. To reduce bias, participants without detectable myocardial fibrosis were entered into an observational registry, such that site investigators were blinded to the presence or absence of myocardial fibrosis in patients randomized to receive guideline-directed conservative management.

The EASY-AS trial (NCT04204915) was launched in the UK in 2020 and randomized all patients with asymptomatic severe aortic stenosis to receive early intervention or guideline-directed conservative management. Patients could be co-enrolled and randomized into both the EVOLVED and EASY-AS trials after cardiac magnetic resonance imaging had been performed as stipulated by the EVOLVED protocol. Treatment randomization was based on randomized treatment assignment in the EASY-AS trial (see eMethods in Supplement 1 for more details).[27]

Trial Intervention

The choice of surgical aortic valve replacement or transcatheter aortic valve implantation was made by the local heart valve team, with the procedure performed as soon as possible (ideally within 2 months) within the constraints of the local health care setting. Patients assigned to receive guideline-directed conservative management received treatment and were referred for aortic valve intervention at the discretion of their treating physician and local heart valve team.

© 2024 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Michael Bowdish on 10/30/2024

Table 1. Clinical Characteristics of the Study Population

	No. (%)	
	Early intervention (n = 113)	Conservative management (n = 111)
Demographics		
Age, median (IQR), y	75 (68-79)	76 (68-80)
Age ≥75 y	57 (50)	60 (54)
Sex		
Male	82 (73)	79 (71)
Female	31 (27)	32 (29)
BMI, median (IQR)	27.2 (24.4-31.1)	27.8 (24.8-31.1)
BMI ≥30	35 (31)	33 (30)
Smoking history (currently or previously smoked)	51 (45)	55 (50)
Comorbidities		
Hypertension	76 (67)	70 (63)
Hyperlipidemia	55 (49)	56 (50)
Diabetes	15 (13)	26 (23)
Cerebrovascular disease	8 (7)	14 (13)
History of angina	5 (4)	8 (7)
Peripheral vascular disease	4 (4)	9 (8)
Past procedures		
Previous percutaneous coronary intervention	7 (6)	7 (6)
Previous coronary artery bypass graft surgery	3 (3)	3 (3)
Medication		
Statin	70 (62)	73 (66)
β-Blocker	33 (29)	17 (15)
Angiotensin-converting enzyme inhibitor	30 (27)	31 (28)
Diuretic	27 (24)	18 (16)
Angiotensin-receptor blocker	25 (22)	21 (19)
High-sensitivity cardiac troponin I concentration, median (IQR), ng/L[a]	11.0 (9.0-18.0)	9.0 (6.0-16.5)
Presence of left ventricular hypertrophy on electrocardiogram	88 (78)	87 (78)
Echocardiography findings, mean (SD)		
Aortic valve peak velocity, m/s	4.3 (0.5)	4.4 (0.5)
Mean gradient, mm Hg	45.2 (11.5)	45.0 (10.2)
Aortic valve area, cm^2	0.8 (0.2)	0.8 (0.2)
Cardiac magnetic resonance findings		
Bicuspid aortic valve	36 (32)	28 (25)
Indexed left ventricular mass, mean (SD), g/m^2	85.5 (18.4)	81.5 (16.4)
Indexed left ventricular end diastolic volume, mean (SD), mL/m^2	75.0 (18.4)	74.3 (18.5)
Indexed left ventricular stroke volume, mean (SD), mL/m^2	50.2 (11.5)	49.9 (11.7)
Left ventricular ejection fraction, mean (SD), %	68 (9)	68 (8)
Prior myocardial infarction	10 (9)	9 (8)

Abbreviation: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared).

[a] High-sensitivity cardiac troponin I concentration ≥6 ng/L was in the inclusion criteria for sites screening with electrocardiogram and troponin.

Trial End Points

The primary end point was a composite of all-cause mortality or unplanned aortic stenosis–related hospitalization during the follow-up period. Aortic stenosis–related hospitalization was defined as any unplanned admission before or after aortic valve replacement with syncope, heart failure, chest pain, ventricular arrythmia, or second- or third-degree heart block attributed to aortic valve disease and adjudicated independently by 2 investigators blinded to the details of trial group randomization. Secondary end points included the individual components of the composite primary end point, symptom burden assessed by the New York Heart Association (NYHA) classification, and the development of left ventricular systolic dysfunction (ejection fraction less than 50%) 12 months after randomization. Health and disability burden were assessed using the World Health Organization Disability Assessment Schedule (WHODAS) 2.0 score 12 months after enrollment. Other prespecified secondary end points included cardiovascular death; aortic stenosis–related death; sudden cardiac death; stroke; endocarditis; implantation of a cardiac pacemaker, resynchronization device, or automated cardioverter defibrillator; and postoperative complications occurring within 30 days.

Statistical Analysis

We estimated that a sample of 356 participants would provide the trial with 80% power at a 2-sided significance level of .05 to detect a significant difference in the primary end point, assuming the incidence of the primary end point would be 25.0% with guideline-directed conservative management and 13.4% with early aortic valve intervention during a follow-up period that continued for a minimum of 12 months after the last patient was enrolled.[13] During the COVID-19 pandemic, recruitment into the EVOLVED trial was suspended for 5 months to comply with a UK government directive. After the pandemic, recruitment rates did not fully recover (eFigure 2 in Supplement 1). The results of 2 emerging randomized trials suggested a larger treatment effect of early intervention than had been assumed in the original power calculation, such that only 35 events would be required to achieve a hazard ratio of 0.33 at 90% power.[6,7] A decision to halt recruitment into the trial on the prespecified date of October 31, 2022, was made by the trial steering committee.

Analyses were performed on an intention-to-treat basis. Ineligible randomized participants were excluded from the primary analysis group (eMethods in Supplement 1). Cox proportional hazard regression was used for analysis of the primary end point and prespecified secondary end points. The NYHA classification was assessed by a proportional odds regression model and the WHODAS 2.0 score was assessed with a linear regression model. All analyses were adjusted for age, sex, and treatment group. Estimates of cumulative incidences were calculated in a time-to-first-event Kaplan-Meier analysis. Because there was no adjustment for multiplicity of testing, secondary end points are reported as point estimates and 95% CIs. The CIs have not been adjusted for multiplicity and should not be used in place of a hypothesis test. All reported *P* values were 2-sided and a value of less than .05 was considered statistically significant. SAS software version 9.4 (SAS Institute) was used for statistical analysis.

© 2024 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Michael Bowdish on 10/30/2024

Results

Between August 4, 2017, and October 31, 2022, a total of 427 asymptomatic patients with severe aortic stenosis were screened, of whom 275 were eligible based on high-sensitivity troponin I concentrations or 12-lead electrocardiogram criteria and underwent cardiac magnetic resonance imaging that was well tolerated and incurred no adverse reactions. Cardiac magnetic resonance imaging identified that 226 of the 275 patients had midwall myocardial fibrosis and were randomized (**Figure 1**). Two participants were excluded because they were randomized after they had been referred for surgery (eMethods in Supplement 1). Of the 224 eligible participants with myocardial fibrosis who had been randomized, 113 patients were randomized to receive early aortic valve intervention and 111 to receive guideline-directed conservative management. Forty-nine patients did not have midwall myocardial fibrosis and were entered into the observational registry. Data collection ended in July 2024, which was 21 months after the last enrolled patient was randomized. Median follow-up was 42 months, with a total follow-up of 722 patient-years.

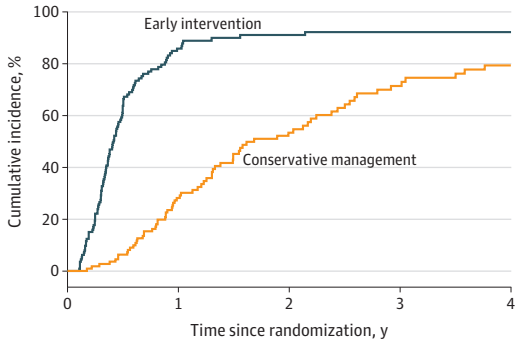
Baseline characteristics were comparable between the groups. The mean (SD) age was 73 (9) years, 63 (28%) were female, and 64 (29%) had a bicuspid aortic valve. Mean (SD) aortic valve peak velocity was 4.3 (0.5) m/s and the mean (SD) aortic valve area was 0.8 (0.2) cm^2 (**Table 1**).

In the early intervention group, 106 patients (94%) received aortic valve intervention and 97 (86%) received it within 12 months of randomization (**Figure 2**). The overall median (IQR) time to intervention was 5.0 (3.4-8.0) months: 4.2 (3.0-6.0) months before the COVID-19 pandemic and 6.5 (4.6-10.9) months after the COVID-19 pandemic. Seven patients (6%) randomized to the early intervention group did not undergo any intervention, of whom 6 died before their procedure at a median (IQR) of 5.5 (1.6-7) months after randomization. Surgical aortic valve replacement was performed in 80 patients (75%) and transcatheter aortic valve intervention in 26 patients (25%) (eTable 3 in Supplement 1). Seven participants (7%) received a mechanical valve and 3 (3%) required urgent inpatient surgery 4.8, 5.6, and 6.1 months after randomization. The 30-day mortality rate was 1%.

In the guideline-directed conservative management group, the median (IQR) time to intervention was 20.2 (11.4-42.0) months: 85 patients (77%) received aortic valve intervention and 31 (28%) received it within 12 months of randomization (Figure 2). Surgical aortic valve replacement was performed in 47 patients (55%) and 38 (45%) underwent transcatheter aortic valve intervention. Symptom development was the primary indication for aortic valve intervention in 61 patients (72%), and 13 patients (15%) required urgent inpatient surgery. Thirty-day mortality was 0%. Additional surgical procedure information is provided in eTable 3 in Supplement 1.

Twenty patients (18%) in the early aortic valve intervention group and 25 (23%) in the guideline-directed conservative management group experienced the primary composite end point of all-cause death or unplanned aortic stenosis hospitalization (hazard ratio, 0.79 [95% CI, 0.44-1.43]; P = .44; between-group difference, −4.82% [95% CI, −15.31% to 5.66%]) (**Table 2** and **Figure 3**).

Figure 2. Cumulative Incidence of Aortic Valve Intervention



No. of patients at risk					
Early intervention	113	16	8	7	7
Conservative management	111	77	37	18	12

At 12 months, 86% of patients in the early intervention group received aortic valve intervention compared with 28% of patients in the guideline-directed conservative management group.

Prespecified secondary end points are listed in Table 2. A total of 16 deaths (14%) occurred in the early intervention group and 14 (13%) in the guideline-directed conservative management group (hazard ratio, 1.22 [95% CI, 0.59-2.51]) (Figure 3). Six of the 16 deaths in the early intervention group and 5 of the 14 deaths in the guideline-directed conservative management group were adjudicated to be related to aortic stenosis (eTable 4 in Supplement 1). Only 1 participant died due to COVID-19 and had been randomized to the early aortic valve intervention group. The frequency of periprocedural complications was low and similar in the 2 groups (eTable 3 in Supplement 1).

Seven patients (6%) in the early aortic valve intervention group and 19 patients (17%) in the guideline-directed conservative management group experienced an unplanned aortic stenosis-related hospitalization (hazard ratio, 0.37 [95% CI, 0.16-0.88]) (Figure 3). At 1 year of follow-up, 21 patients (20%) in the early intervention group and 39 (38%) in the guideline-directed conservative management group had NYHA class II-IV symptoms (adjusted odds ratio, 0.37 [95% CI, 0.20-0.70]) (eFigure 3 in Supplement 1). The adjusted mean WHODAS 2.0 score at 1 year was 3.3 in patients in the early intervention group compared with 4.1 in the guideline-directed conservative management group (adjusted mean difference, −0.8 [95% CI, −2.0 to 0.4]).

Discussion

In this study that compared early aortic valve intervention with guideline-directed conservative management in asymptomatic patients with severe aortic stenosis and subclinical evidence of cardiac decompensation, there was no demonstrable difference in the primary composite end point of all-cause mortality or unplanned aortic stenosis-related hospitalization. However, the 95% CI around the primary end point

© 2024 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Michael Bowdish on 10/30/2024

Table 2. Primary and Secondary End Points

Outcome[a]	No. (%) Early intervention (n = 113)	No. (%) Conservative management (n = 111)	Absolute difference (95% CI), %	Hazard ratio (95% CI)
Primary end point				
All-cause death or unplanned aortic stenosis-related hospitalization	20 (18)	25 (23)	−4.82 (−15.31 to 5.66) [*P* = .37]	0.79 (0.44 to 1.43) [*P* = .44]
Secondary end points				
All-cause death	16 (14)	14 (13)	1.55 (−7.37 to 10.46)	1.22 (0.59 to 2.51)
Cardiovascular death	10 (9)	8 (7)	1.64 (−5.47 to 8.75)	1.33 (0.52 to 3.36)
Aortic stenosis-related death	6 (5)	5 (5)	0.81 (−4.85 to 6.46)	1.25 (0.38 to 4.10)
Unplanned aortic stenosis-related hospitalization	7 (6)	19 (17)	−10.92 (−19.22 to 2.62)	0.37 (0.16 to 0.88)
Permanent pacemaker, cardiac resynchronization therapy, or automated cardiac defibrillator implantation	5 (4)	7 (6)	−1.88 (−7.78 to 4.02)	0.75 (0.24 to 2.37)
Stroke	8 (7)	14 (13)	−5.53 (−13.31 to 2.25)	0.62 (0.26 to 1.49)
Endocarditis	1 (1)	3 (3)	−1.82 (−5.29 to 1.66)	0.33 (0.03 to 3.14)
Development of left ventricular systolic impairment	8 (7)	11 (10)	−2.83 (−10.13 to 4.47)	0.72 (0.29 to 1.80)
Perioperative or postoperative complications within 30 d of surgery or transcatheter aortic valve intervention	15 (14)	9 (11)	5.17 (−2.89 to 13.22)	Odds ratio for ≥1 specified complication, 1.20 (95% CI, 0.50 to 2.93)
WHODAS 2.0 score at 1 y, adjusted mean[b]	3.3	4.1		Adjusted mean difference, −0.8 (95% CI, −2.0 to 0.4)
NYHA classification at 1 y				Odds ratio for higher NYHA classification, 0.37 (95% CI, 0.20 to 0.70)
Class I (least severe)	86 (80)	64 (62)		
Class II	19 (18)	30 (29)		
Class III	2 (2)	8 (8)		
Class IV (most severe)	0	1 (1)		

Abbreviations: NYHA, New York Heart Association; WHODAS, World Health Organization Disability Assessment Schedule.

[a] Cardiovascular death is defined as death due to myocardial infarction, sudden cardiac death, heart failure, stroke, or other cardiovascular causes; death due to cardiovascular procedures; and death due to other cardiovascular causes. Aortic stenosis–related death is a death in which aortic stenosis has been listed as a contributory cause by the clinical care team on the patient's official death certificate. Sudden cardiac death is defined as any death that occurs unexpectedly and not within 30 days of acute myocardial infarction. This includes unsuccessful resuscitation following an arrhythmia.

is wide and encompasses potential clinically meaningful benefits or harms from early intervention. The findings are not definitive, and further research will be required to confirm the trial findings.

In this trial, the population for increased cardiac risk was enriched by using cardiac biomarkers and cardiac magnetic resonance imaging, and a patient population was selected with aortic stenosis in whom the left ventricle was starting to decompensate due to their severe valve disease. The hypothesis was that these high-risk asymptomatic patients would have the most to gain from a strategy of earlier aortic valve intervention. Despite this, an impact of the trial intervention on the primary outcome was still not demonstrated.

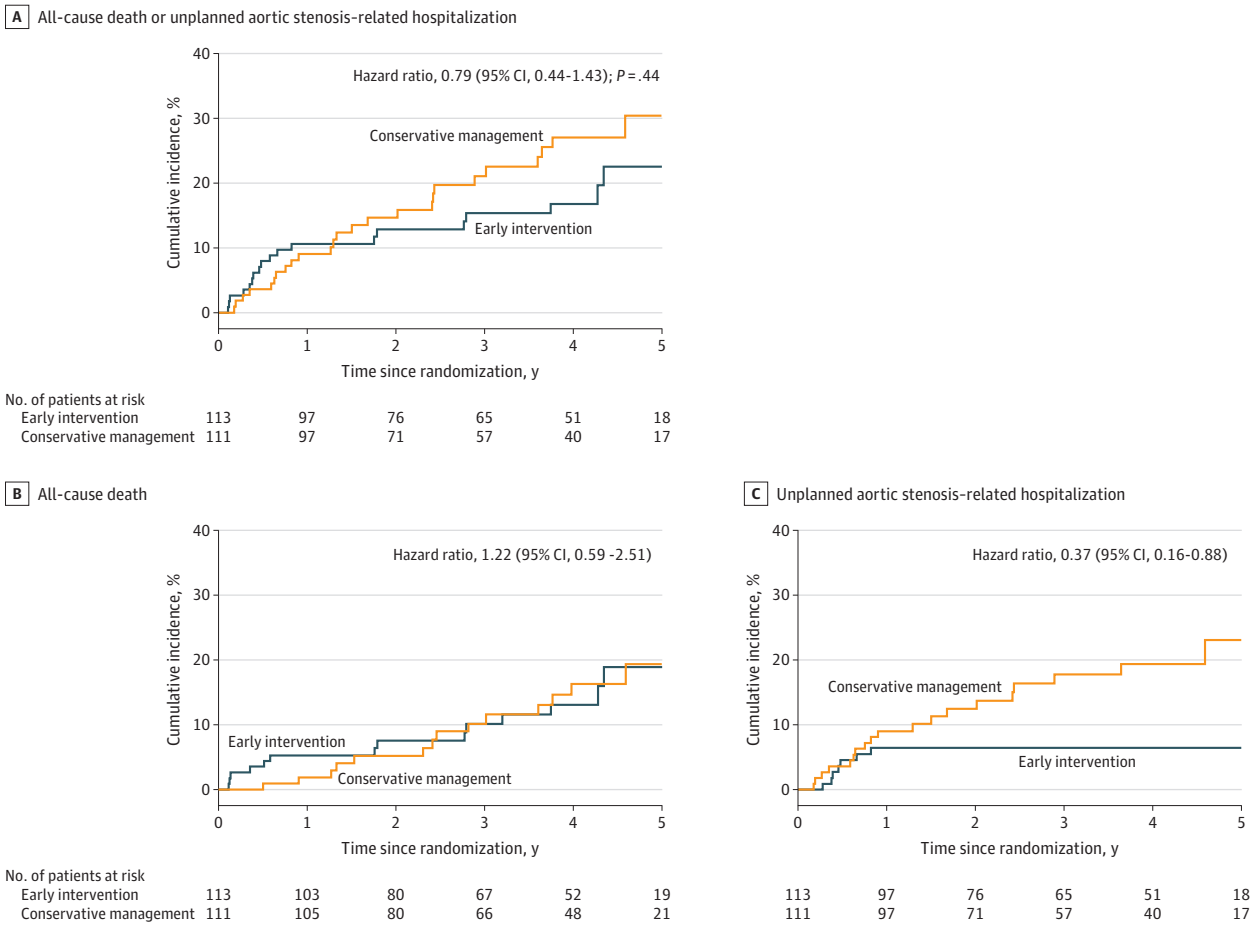
It could be argued that the median time to early intervention was too long in the early intervention group and that some patients may not have had a primary outcome event had they undergone more rapid early intervention. However, the time delay to intervention is representative of contemporary practice in the UK and Australia and indeed many other health care systems around the world, including Canada, France, and Sweden.[28-31] Nonetheless, there was a difference of 15 months in the time to intervention between trial groups. This occurred despite a marginally shorter time to intervention from referral in the guideline-directed conservative management group, which was likely driven by the development of symptoms and the higher rates of hospitalization and inpatient procedures.

The decision to undertake aortic valve intervention in an asymptomatic patient requires careful consideration, because the early procedural risks need to be weighed against those associated with progressive and potentially irreversible adverse left ventricular remodeling, heart failure, and death with continued conservative management. International guidelines suggest that conservative management and watchful waiting for the onset of symptoms is a safe strategy, and this is supported by the data in this trial. The risk of procedural death and sudden cardiac death were very low even in the enriched older population of patients with severe aortic stenosis and myocardial fibrosis. Moreover, there were no differences in all-cause or cardiovascular mortality between trial groups across 722 patient-years of follow-up.

© 2024 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Michael Bowdish on 10/30/2024

Figure 3. Cumulative Incidence of the Primary Composite End Point and Its Components



Time-to-first-event Kaplan-Meier analysis. The median (IQR) observation times were (A) 48.2 (12.6-52.1) months for early intervention and 36.5 (13.8-49.6) months for conservative management; (B) 48.5 (15.9-52.9) months for early intervention and 48.1 (23.7-55.3) months for conservative management; and (C) 48.2 (12.6-52.1) months for early intervention and 36.5 (13.8-49.6) months for conservative management.

The mortality data of this study conflict with the results of the RECOVERY trial, which demonstrated a mortality benefit of early intervention in a highly select cohort of younger and otherwise healthy patients with predominantly bicuspid valve disease. These patients had critical aortic stenosis with a mean aortic valve peak velocity of more than 5 m/s, meaning most would have met the Class IIb level recommendation for aortic valve intervention at trial inclusion.[2,3,6] The AVATAR trial demonstrated a long-term benefit in all-cause, but not cardiovascular, mortality, but also recruited a younger patient population that included patients with very severe aortic stenosis. Even after enriching for a high-risk population with myocardial fibrosis, the current trial suggests that their findings of improved mortality with early intervention cannot be extrapolated to the broader older population with asymptomatic severe aortic stenosis who have a greater burden of comorbidities. Indeed, in the current study, only one-third of deaths were attributed to the patients' underlying aortic valve disease, meaning that most deaths were not modifiable by aortic valve intervention.

Although focus on mortality is important, it is also crucial to consider the impact of intervention in reducing symptoms and preventing emergency hospitalizations in an older population. The current study showed a higher burden of heart failure symptoms at 12 months in patients receiving guideline-directed conservative management that was not apparent in those who underwent early aortic valve intervention. Consistent with this, early aortic valve intervention also resulted in fewer unplanned aortic stenosis-related hospitalizations compared with guideline-directed conservative management. These are primary treatment goals for many older patients and thus represent an important finding, particularly given the low periprocedural risk associated with early intervention.[6,7] The findings that early intervention was associated with a lower incidence of unplanned aortic stenosis-related hospitalization and improved symptom burden should be considered hypothesis-generating given that the study did not meet the primary end point and was not adjusted for multiple comparisons. If confirmed, the principal benefits of early intervention in asymptomatic patients with subclinical cardiac

© 2024 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Michael Bowdish on 10/30/2024

decompensation may be to avoid the development of symptoms and unplanned hospitalizations rather than to reduce mortality. These hypotheses need to be addressed in future larger trials, such as the EASY-AS trial (NCT04204915).[27]

Limitations

This trial has several limitations. First, because the primary end point is null, any conclusions about the secondary end points must be designated as hypothesis-generating. Second, recruitment was heavily impacted by the COVID-19 pandemic and prevented achievement of the original sample size. For these reasons, further research is needed to confirm the findings. Third, the rate of transcatheter aortic valve intervention was higher in the guideline-directed conservative management group than the early intervention group, reflecting better access to transcatheter aortic valve intervention during study conduct and urgent intervention following unplanned aortic stenosis-related hospitalizations, where patients may have been too unwell to undergo surgical aortic valve replacement. Fourth, the percentage of female participants in this trial was low (28%), which could reflect that female patients may have less advanced myocardial remodeling than males in response to the same level of valvular stenosis,[32,33] and this limits the generalizability of the trial findings.

Conclusions

Early aortic valve intervention has no demonstrable effect on the combined primary endpoint of all-cause death or unplanned aortic stenosis-related hospitalization compared with guideline-directed conservative management among patients with asymptomatic severe aortic stenosis and myocardial fibrosis. There was a wide 95% CI around the primary end point, with further research needed to confirm these findings.

ARTICLE INFORMATION

Accepted for Publication: October 10, 2024.

Published Online: October 28, 2024. doi:10.1001/jama.2024.22730

Author Affiliations: British Heart Foundation Centre of Research Excellence, University of Edinburgh, Edinburgh, Scotland (Loganath, Craig, White, Tuck, Mills, Newby, Dweck); Edinburgh Heart Centre, Royal Infirmary of Edinburgh, Edinburgh, Scotland (Craig, Everett, Bing, White, Mills, Newby, Dweck); Norwich Medical School, University of East Anglia, Norwich, United Kingdom (Tsampasian, Vassiliou); Department of Cardiology, Norfolk and Norwich University Hospital, Norwich, United Kingdom (Tsampasian, Vassiliou); Department of Coronary Disease and Heart Failure, Jagiellonian University Medical College, Krakow, Poland (Molek); University of Medicine and Pharmacy Carol Davila, Cardiology Department, Euroecholab, Bucharest, Romania (Botezatu); Department of Cardiovascular Sciences, University of Leicester and the NIHR Leicester Biomedical Research Centre, Glenfield Hospital, Leicester, United Kingdom (Aslam, McCann, Singh); Edinburgh Clinical Trials Unit, Usher Institute, The University of Edinburgh, Edinburgh, Scotland (Lewis, Rayson, Cranley, Irvine, Armstrong, Milne); Edinburgh Clinical Research Facility, The University of Edinburgh, Edinburgh, Scotland (Graham); Edinburgh Imaging, The University of Edinburgh, Edinburgh, Scotland (MacGillivray); Department of Cardiology, National Heart Centre Singapore, Singapore (Chin); Cardiovascular Medicine ACP Duke NUS Medical School, Singapore (Chin); Medical School, University of Western Australia, Perth, Western Australia, Australia (Hillis); Department of Cardiology, Royal Perth Hospital, Perth, Western Australia, Australia (Hillis); Department of Cardiology, Liverpool Centre for Cardiovascular Science, Liverpool Heart and Chest Hospital, Liverpool, United Kingdom (Fairbairn); Leeds Teaching Hospitals NHS Trust, Leeds, United Kingdom (Greenwood); Baker Heart and Diabetes Institute, Melbourne, Victoria, Australia (Greenwood); Department of Cardiology, Queen Elizabeth Hospital Birmingham, University Hospitals of Birmingham NHS Foundation Trust, Birmingham, United Kingdom (Steeds); Cardiac Unit, Raigmore Hospital, Inverness, Scotland (Leslie); Division of Molecular and Clinical Medicine, School of Medicine, University of Dundee, Dundee, Scotland, United Kingdom (Lang); Tuanku Muhriz Chair, National University of Malaysia, Malaysia (Lang); Bristol Heart Institute, University Hospitals Bristol NHS Foundation Trust, Bristol, United Kingdom (Bucciarelli-Ducci, Joshi); Royal Brompton and Harefield Hospitals, Guys' and St Thomas NHS Foundation Trust, London, United Kingdom (Bucciarelli-Ducci, Prasad); Translational and Clinical Research Institute, Faculty of Medical Sciences, Newcastle, United Kingdom (Kunadian); University and Cardiothoracic Centre, Freeman Hospital, Newcastle upon Tyne Hospitals NHS Foundation Trust, Newcastle upon Tyne, United Kingdom (Kunadian); Essex Cardiothoracic Centre, Nethermayne, Basildon, Essex, United Kingdom (Dungu); Anglia Ruskin University, Chelmsford, Essex, United Kingdom (Dungu); Department of Cardiology, Royal Wolverhampton NHS Trust, Wolverhampton, UK, Institute of Cardiovascular Sciences, University of Birmingham, United Kingdom (Hothi); Retired, Hertfordshire, United Kingdom (Boon); Department of Cardiology, West Hertfordshire Hospitals NHS Trust, Watford, United Kingdom (Keenan); Imperial College, London, United Kingdom (Keenan); Aberdeen Cardiovascular and Diabetes Centre, University of Aberdeen, Aberdeen, Scotland (Dawson); Institute of Cardiovascular Sciences, University College London, and St Bartholomew's Hospital, Barts Health NHS Trust, West Smithfield, London, United Kingdom (Treibel); Department of Cardiology, Manchester Heart Institute, Manchester Royal Infirmary, Manchester University NHS Foundation Trust, Manchester, United Kingdom (Motwani); Division of Cardiovascular Sciences, School of Medical Sciences, Faculty of Biology, Medicine and Health, Manchester Academic Health Science Centre, The University of Manchester, Manchester, United Kingdom (Miller); Usher Institute, The University of Edinburgh, Edinburgh, Scotland (Mills); Guy's and St Thomas' NHS Foundation Trust, London, United Kingdom (Rajani); Cardiology, Northumbria Healthcare NHS Foundation Trust, Newcastle upon Tyne, United Kingdom (Ripley); Cleveland Clinic London and St Thomas' Hospital, London, United Kingdom (Prendergast).

Author Contributions: Drs Dweck and Lewis had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Drs Loganath, Craig, Everett, and Bing contributed equally.
Concept and design: Everett, Tuck, Rayson, Cranley, Irvine, Kunadian, Vassiliou, Dungu, Hothi, Boon, McCann, Newby, Dweck.
Acquisition, analysis, or interpretation of data: Loganath, Craig, Everett, Bing, Tsampasian, Molek, Botezatu, Aslam, Lewis, Graham, White, MacGillivray, Tuck, Rayson, Cranley, Irvine, Milne, Chin, Hillis, Fairbairn, Greenwood, Steeds, Leslie, Lang, Bucciarelli-Ducci, Kunadian, Vassiliou, Dungu, Keenan, Dawson, Treibel, Motwani, Miller, Mills, Rajani, Ripley, McCann, Prendergast, Singh, Newby, Dweck.
Drafting of the manuscript: Loganath, Craig, Graham, Cranley, Irvine, Kunadian, Dungu, Keenan, Dawson, Ripley, Dweck.
Critical review of the manuscript for important intellectual content: Loganath, Craig, Everett, Bing, Tsampasian, Molek, Botezatu, Aslam, Lewis, Graham, White, MacGillivray, Tuck, Rayson, Milne, Chin, Hillis, Fairbairn, Greenwood, Steeds, Leslie, Lang, Bucciarelli-Ducci, Kunadian, Vassiliou, Dungu, Hothi, Boon, Keenan, Dawson, Treibel, Motwani, Miller, Mills, Rajani, Ripley, McCann, Prendergast, Singh, Newby, Dweck.
Statistical analysis: Loganath, Craig, Lewis, Graham.
Obtained funding: Cranley, Irvine, Mills, Newby, Dweck.
Administrative, technical, or material support: Everett, Molek, White, MacGillivray, Tuck, Rayson, Cranley, Irvine, Milne, Chin, Hillis, Steeds, Dungu, Hothi, Boon, Dawson, Treibel, Motwani, Miller, Rajani, Newby, Dweck.
Supervision: Loganath, Lewis, White, Greenwood, Leslie, Bucciarelli-Ducci, Vassiliou, Dungu, Hothi, Boon, Dawson, Motwani, Ripley, Prendergast, Newby, Dweck.
Other - Local primary investigator and member of trial steering committee: McCann.
Other - Trial recruitment: Rajani.
Other - Data curation: Molek.
Other - Patient recruitment: Kunadian.

Conflict of Interest Disclosures: Dr Lang reported receiving grants from Novo Nordisk, Boehringer

© 2024 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Michael Bowdish on 10/30/2024

Ingelheim, AstraZeneca, Moderna, and Roche Diagnostics outside the submitted work. Dr Bucciarelli-Ducci reported receiving personal fees from Siemens Healthineers, GE HealthCare, Philips, and Bayer outside the submitted work and serving as chief executive officer (part-time contractor) for the Society for Cardiovascular Magnetic Resonance. Dr Keenan reported shareholding and nonfinancial support from MyCardium AI outside the submitted work. Dr Treibel reported receiving personal fees from AstraZeneca outside the submitted work. Dr Miller reported receiving nonfinancial support from Univar Solutions and grants from National Institute for Health and Care Research, Amicus Therapeutics, Guerbet Laboratories Limited, and AstraZeneca and personal fees from AstraZeneca, PureTech Health, Novo Nordisk, Boehringer Ingelheim, HAYA Therapeutics, and Eli Lilly outside the submitted work. Dr McCann reported receiving grants from University of Edinburgh during the conduct of the study and grants from British Heart Foundation for another trial of intervention in aortic stenosis outside the submitted work. Dr Prendergast reported receiving personal fees from Edwards Lifesciences, Valvosoft, and Medtronic outside the submitted work. Dr Dweck reported receiving personal fees from Edwards outside the submitted work and receiving speaker fees from Novartis and consultancy fees from Novartis, Jupiter Bioventures, AstraZeneca, Beren Therapeutics, and Silence Therapeutics. No other disclosures were reported.

Funding/Support: Sir Jules Thorne Charitable Trust funded the trial.

Role of the Funder/Sponsor: Sir Jules Thorne Charitable Trust had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 4.

REFERENCES

1. Osnabrugge RL, Mylotte D, Head SJ, et al. Aortic stenosis in the elderly: disease prevalence and number of candidates for transcatheter aortic valve replacement: a meta-analysis and modeling study. *J Am Coll Cardiol*. 2013;62(11):1002-1012. doi:10.1016/j.jacc.2013.05.015

2. Vahanian A, Beyersdorf F, Praz F, et al; ESC/EACTS Scientific Document Group. 2021 ESC/EACTS guidelines for the management of valvular heart disease. *Eur Heart J*. 2022;43(7):561-632.

3. Otto CM, Nishimura RA, Bonow RO, et al. 2020 ACC/AHA guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2021;143(5):e72-e227.

4. Zilberszac R, Gabriel H, Schemper M, Laufer G, Maurer G, Rosenhek R. Asymptomatic severe aortic stenosis in the elderly. *JACC Cardiovasc Imaging*. 2017;10(1):43-50. doi:10.1016/j.jcmg.2016.05.015

5. Iung B, Cachier A, Baron G, et al. Decision-making in elderly patients with severe aortic stenosis: why are so many denied surgery? *Eur Heart J*. 2005;26(24):2714-2720. doi:10.1093/eurheartj/ehi471

6. Kang DH, Park SJ, Lee SA, et al. Early surgery or conservative care for asymptomatic aortic stenosis. *N Engl J Med*. 2020;382(2):111-119. doi:10.1056/NEJMoa1912846

7. Banovic M, Putnik S, Penicka M, et al; AVATAR Trial Investigators. Aortic valve replacement versus conservative treatment in asymptomatic severe aortic stenosis: the AVATAR Trial. *Circulation*. 2022;145(9):648-658. doi:10.1161/CIRCULATIONAHA.121.057639

8. Shah AS, Griffiths M, Lee KK, et al. High sensitivity cardiac troponin and the under-diagnosis of myocardial infarction in women: prospective cohort study. *BMJ*. 2015;350:g7873. doi:10.1136/bmj.g7873

9. Chin CW, Shah AS, McAllister DA, et al. High-sensitivity troponin I concentrations are a marker of an advanced hypertrophic response and adverse outcomes in patients with aortic stenosis. *Eur Heart J*. 2014;35(34):2312-2321. doi:10.1093/eurheartj/ehu189

10. Shah AS, Chin CW, Vassiliou V, et al. Left ventricular hypertrophy with strain and aortic stenosis. *Circulation*. 2014;130(18):1607-1616. doi:10.1161/CIRCULATIONAHA.114.011085

11. Greve AM, Boman K, Gohlke-Baerwolf C, et al. Clinical implications of electrocardiographic left ventricular strain and hypertrophy in asymptomatic patients with aortic stenosis: the Simvastatin and Ezetimibe in Aortic Stenosis study. *Circulation*. 2012;125(2):346-353. doi:10.1161/CIRCULATIONAHA.111.049759

12. Azevedo CF, Nigri M, Higuchi ML, et al. Prognostic significance of myocardial fibrosis quantification by histopathology and magnetic resonance imaging in patients with severe aortic valve disease. *J Am Coll Cardiol*. 2010;56(4):278-287. doi:10.1016/j.jacc.2009.12.074

13. Dweck MR, Joshi S, Murigu T, et al. Midwall fibrosis is an independent predictor of mortality in patients with aortic stenosis. *J Am Coll Cardiol*. 2011;58(12):1271-1279. doi:10.1016/j.jacc.2011.03.064

14. Vassiliou VS, Perperoglou A, Raphael CE, et al. Midwall fibrosis and 5-year outcome in moderate and severe aortic stenosis. *J Am Coll Cardiol*. 2017;69(13):1755-1756. doi:10.1016/j.jacc.2017.01.034

15. Dweck MR, Boon NA, Newby DE. Calcific aortic stenosis: a disease of the valve and the myocardium. *J Am Coll Cardiol*. 2012;60(19):1854-1863. doi:10.1016/j.jacc.2012.02.093

16. Hein S, Arnon E, Kostin S, et al. Progression from compensated hypertrophy to failure in the pressure-overloaded human heart: structural deterioration and compensatory mechanisms. *Circulation*. 2003;107(7):984-991. doi:10.1161/01.CIR.0000051865.66123.B7

17. Krayenbuehl HP, Hess OM, Monrad ES, Schneider J, Mall G, Turina M. Left ventricular myocardial structure in aortic valve disease before, intermediate, and late after aortic valve replacement. *Circulation*. 1989;79(4):744-755. doi:10.1161/01.CIR.79.4.744

18. Kong WKF, Bax JJ, Delgado V. Left ventricular myocardial fibrosis: a marker of bad prognosis in symptomatic severe aortic stenosis. *Eur Heart J*. 2020;41(20):1915-1917. doi:10.1093/eurheartj/ehaa151

19. Milano AD, Faggian G, Dodonov M, et al. Prognostic value of myocardial fibrosis in patients with severe aortic valve stenosis. *J Thorac Cardiovasc Surg*. 2012;144(4):830-837. doi:10.1016/j.jtcvs.2011.11.024

20. Barone-Rochette G, Piérard S, De Meester de Ravenstein C, et al. Prognostic significance of LGE by CMR in aortic stenosis patients undergoing valve replacement. *J Am Coll Cardiol*. 2014;64(2):144-154. doi:10.1016/j.jacc.2014.02.612

21. Chin CWL, Everett RJ, Kwiecinski J, et al. Myocardial fibrosis and cardiac decompensation in aortic stenosis. *JACC Cardiovasc Imaging*. 2017;10(11):1320-1333. doi:10.1016/j.jcmg.2016.10.007

22. Musa TA, Treibel TA, Vassiliou VS, et al. Myocardial scar and mortality in severe aortic stenosis. *Circulation*. 2018;138(18):1935-1947. doi:10.1161/CIRCULATIONAHA.117.032839

23. Thornton GD, Vassiliou VS, Musa TA, et al; BSCMR AS700 Consortium. Myocardial scar and remodelling predict long-term mortality in severe aortic stenosis beyond 10 years. *Eur Heart J*. 2024;45(22):2019-2022. doi:10.1093/eurheartj/ehae067

24. Bing R, Everett RJ, Tuck C, et al. Rationale and design of the randomized, controlled Early Valve Replacement Guided by Biomarkers of Left Ventricular Decompensation in Asymptomatic Patients with Severe Aortic Stenosis (EVOLVED) trial. *Am Heart J*. 2019;212:91-100. doi:10.1016/j.ahj.2019.02.018

25. Schulz KF, Altman DG, Moher D; CONSORT Group. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. *BMJ*. 2010;340:c332. doi:10.1136/bmj.c332

26. Baumgartner H, Hung J, Bermejo J, et al; American Society of Echocardiography; European Association of Echocardiography. Echocardiographic assessment of valve stenosis: EAE/ASE recommendations for clinical practice. *J Am Soc Echocardiogr*. 2009;22(1):1-23. doi:10.1016/j.echo.2008.11.029

27. Richardson C, Gilbert T, Aslam S, et al. Rationale and design of the early valve replacement in severe asymptomatic aortic stenosis trial. *Am Heart J*. 2024;275:119-127. doi:10.1016/j.ahj.2024.05.013

28. Elbaz-Greener G, Masih S, Fang J, et al. Temporal trends and clinical consequences of wait times for transcatheter aortic valve replacement: a population-based study. *Circulation*. 2018;138(5):483-493. doi:10.1161/CIRCULATIONAHA.117.033432

29. Henning KA, Ravindran M, Qiu F, et al. Impact of procedural capacity on transcatheter aortic valve replacement wait times and outcomes: a study of regional variation in Ontario, Canada. *Open Heart*. 2020;7(1):e001241. doi:10.1136/openhrt-2020-001241

30. Nilsson K, Lindholm D, Backes J, et al. Regional assessment of availability for transcatheter aortic valve implantation in Sweden: a long-term observational study. *Eur Heart J Qual Care Clin Outcomes*. Published December 29, 2023. doi:10.1093/ehjqcco/qcad076

31. Roule V, Rebouh I, Lemaitre A, et al. Impact of wait times on late postprocedural mortality after successful transcatheter aortic valve replacement. *Sci Rep*. 2022;12(1):5967. doi:10.1038/s41598-022-09995-z

32. Treibel TA, Kozor R, Fontana M, et al. Sex dimorphism in the myocardial response to aortic stenosis. *JACC Cardiovasc Imaging*. 2018;11(7):962-973. doi:10.1016/j.jcmg.2017.08.025

33. Singh A, Musa TA, Treibel TA, et al. Sex differences in left ventricular remodelling, myocardial fibrosis and mortality after aortic valve replacement. *Heart*. 2019;105(23):1818-1824. doi:10.1136/heartjnl-2019-314987

© 2024 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Michael Bowdish on 10/30/2024